
TRAVAUX DIRIG  S N   3 : Linear Separators

Stephan CL  MEN  ON <stephan.clemencon@telecom-paris.fr>
Quentin BOUNIOT <quentin.bouniot@telecom-paris.fr>

EXERCICE 1. We work in the framework of binary classification : let X be a random descriptor taking values in $\mathbb{R}$, equipped with its Borel sigma-algebra, and let Y be a random label taking the value 0 or 1. Let $\mathcal{G} := \{g : \mathbb{R} \rightarrow \{0, 1\}\}$ be the set of classifiers suited to this setting. The classification error is defined as the map $L : g \in \mathcal{G} \mapsto \mathbb{P}(Y \neq g(X)) \in [0, 1]$, and we write $L^* := \inf_{g \in \mathcal{G}} L(g)$.

In this exercise, we consider the family $\mathcal{G}_0$ of linear classifiers on $\mathbb{R}$ of the form :

$$g_{(x_0, y_0)} : x \in \mathbb{R} \mapsto \begin{cases} y_0 & \text{if } x \leq x_0, \\ 1 - y_0 & \text{otherwise,} \end{cases}$$

where $(x_0, y_0) \in \mathbb{R} \times \{0, 1\}$. The classification error of such a $g_{(x_0, y_0)}$ is denoted more simply by $L(x_0, y_0)$, and we set $L_0 := \inf_{(x_0, y_0)} L(x_0, y_0)$.

1) Express the classification error of an arbitrary element of $\mathcal{G}_0$ in terms of the conditional distributions of X given Y . Use the notation $F_y(x) := \mathbb{P}\{X \leq x \mid Y = y\}$ for $(x, y) \in \mathbb{R} \times \{0, 1\}$ and $p := \mathbb{P}(Y = 1)$.

2) By considering the points $(x_0, y_0) = (-\infty, 0)$ and $(x_0, y_0) = (-\infty, 1)$, show that $L_0 \leq \frac{1}{2}$.

3) Show that $L_0 = \frac{1}{2} - \sup_x \left| p F_1(x) - (1 - p) F_0(x) - p + \frac{1}{2} \right|$. Simplify the expression when $p = \frac{1}{2}$.

Hint. For every $(a, b) \in \mathbb{R}^2$, one can write $\min(a, b) = \frac{a + b - |a - b|}{2}$.

4) Show that $L_0 = \frac{1}{2}$ if and only if $L^* = \frac{1}{2}$.

5) Prove the Chebyshev–Cantelli inequality : for every real-valued random variable Z (with non-zero variance) and every $t \geq 0$,

$$\mathbb{P}(Z - \mathbb{E}(Z) \geq t) \leq \frac{\mathbb{V}(Z)}{\mathbb{V}(Z) + t^2}.$$

6) Let m_y and σ_y^2 denote respectively the expectation and variance of the conditional distribution of X given $Y = y$, with $y \in \{0, 1\}$ and $\sigma_y^2 > 0$. Show that :

$$L_0 \leq \left(1 + \frac{(m_0 - m_1)^2}{(\sigma_0 + \sigma_1)^2} \right)^{-1}.$$

Hint. Use the inequality proved in the previous question.

7) Discuss the performance of the empirical minimizer taken over the class $\mathcal{G}_0$ and the limitations of linear classifiers.

Correction Exercise 1

1) Let $(x, y) \in \mathbb{R} \times \{0, 1\}$. We have :

$$\begin{aligned} L(x, y) &= \mathbb{P}\{Y \neq g_{(x,y)}(X)\} \\ &= \mathbb{P}(X > x | Y = y) \mathbb{P}(Y = y) + \mathbb{P}(X \leq x | Y = 1 - y) \mathbb{P}(Y = 1 - y) \end{aligned}$$

With $\mathbb{P}(X > x | Y = y) = 1 - F_y(x)$ and $\mathbb{P}(X \leq x | Y = 1 - y) = F_{1-y}(x)$, we can separate the two cases $y = 0$ and $y = 1$ to get :

$$\begin{aligned} L(x, 0) &= (1 - F_0(x))(1 - p) + F_1(x)p \\ L(x, 1) &= (1 - F_1(x))p + F_0(x)(1 - p) \end{aligned}$$

Combining the two cases, we can write :

$$\begin{aligned} L(x, y) &= \mathbb{1}_{\{y=0\}} [(1 - F_0(x))(1 - p) + F_1(x)p] + \mathbb{1}_{\{y=1\}} [(1 - F_1(x))p + F_0(x)(1 - p)] \\ &= y [p - pF_1(x) + F_0(x) - pF_0(x)] + (1 - y) [1 - p - F_0(x) + pF_0(x) + pF_1(x)] = 1 - y + (2y - 1)h(x) \end{aligned}$$

With $h(x) = p - pF_1(x) + F_0(x) - pF_0(x)$.

2) Let $y \in \{0, 1\}$. We have : $\lim_{x \rightarrow \infty} F_y(x) = 0$, and $\lim_{x \rightarrow -\infty} h(x) = p$. Therefore, we have :

$$\lim_{x \rightarrow -\infty} L(x, y) = 1 - y + (2y - 1)p = \begin{cases} 1 - p & \text{if } y = 0 \\ p & \text{if } y = 1 \end{cases}$$

Thus, we have $L_0 \leq \min(p, 1 - p) \leq \frac{1}{2}$.

3) By definition :

$$\begin{aligned} L_0 &= \inf_{(x,y) \in \mathbb{R} \times \{0,1\}} L(x, y) \\ &= \inf_{x \in \mathbb{R}} \min(L(x, 0), L(x, 1)) \\ &= \inf_{x \in \mathbb{R}} \min(h(x), 1 - h(x)) \\ &= \inf_{x \in \mathbb{R}} \frac{h(x) + (1 - h(x)) - |h(x) - (1 - h(x))|}{2} \\ &= \inf_{x \in \mathbb{R}} \frac{1}{2} - \left| h(x) - \frac{1}{2} \right| \\ &= \frac{1}{2} - \sup_{x \in \mathbb{R}} \left| h(x) - \frac{1}{2} \right| \end{aligned}$$

Then replacing the value of $h(x)$, we get what we want.

When $p = \frac{1}{2}$, we have $h(x) = \frac{1}{2} + \frac{F_0(x) - F_1(x)}{2}$, and thus :

$$\begin{aligned} L_0 &= \frac{1}{2} - \sup_{x \in \mathbb{R}} \left| \frac{F_0(x) - F_1(x)}{2} \right| \\ &= \frac{1}{2} - \frac{1}{2} \sup_{x \in \mathbb{R}} |F_0(x) - F_1(x)| \\ &= \frac{1}{2} (1 - d_{TV}(F_0, F_1)) \end{aligned}$$

4) First, let's assume $L^* = \frac{1}{2}$. By definition $\mathcal{G}_0 \subset \mathcal{G}$, so $L^* \leq L_0$. However, from question 2, we have $L_0 \leq \frac{1}{2}$. Therefore, we have $L_0 = \frac{1}{2}$.

Now, let's assume $L_0 = \frac{1}{2}$. From question 2 :

$$L_0 = \frac{1}{2} \leq \min(p, 1 - p) \leq \frac{1}{2}$$

Therefore, we have $p = \frac{1}{2}$. Then, from question 3 :

$$\begin{aligned} L_0 &= \frac{1}{2}(1 - d_{TV}(F_0, F_1)) = \frac{1}{2} \\ \iff d_{TV}(F_0, F_1) &= 0 \\ \iff F_0 &= F_1 \\ \iff X|Y = 1 &\sim X|Y = 0 \end{aligned}$$

Then, for any classifier $g \in \mathcal{G}$, we have :

$$\begin{aligned} L(g) &= \mathbb{P}(Y \neq g(X)) \\ &= \mathbb{P}(Y = 1)\mathbb{P}(g(X) = 0|Y = 1) + \mathbb{P}(Y = 0)\mathbb{P}(g(X) = 1|Y = 0) \\ &= \frac{1}{2}\mathbb{P}(g(X) = 0|Y = 1) + \frac{1}{2}\mathbb{P}(g(X) = 1|Y = 0) \quad (\text{since } p = \frac{1}{2}) \\ &= \frac{1}{2}\mathbb{P}(g(X) = 0|Y = 1) + \frac{1}{2}\mathbb{P}(g(X) = 1|Y = 1) \quad (\text{since } X|Y = 1 \sim X|Y = 0) \\ &= \frac{1}{2} \end{aligned}$$

This is true for any classifier $g \in \mathcal{G}$, so we have $L^* = \frac{1}{2}$ (true in inf, and for Bayes classifier).

Remark : We can note that the last part is true as long as $X|Y = 1 \sim X|Y = 0$ (no difference in the data between labels), and $p = \frac{1}{2}$ (probability of each label is the same).

5) Let us note $Z^* = Z - \mathbb{E}(Z)$ (centered version of Z). Then we have :

$$t = \mathbb{E}(t - Z^*), \quad \text{because } \mathbb{E}(Z^*) = 0.$$

Then, we have :

$$\begin{aligned} \mathbb{E}(t - Z^*) &= \mathbb{E}[(t - Z^*)\mathbb{1}_{\{Z^* < t\}}] + \mathbb{E}[(t - Z^*)\mathbb{1}_{\{Z^* \geq t\}}] \\ &\leq \mathbb{E}[(t - Z^*)\mathbb{1}_{\{Z^* < t\}}] \quad (\text{since } (t - Z^*) \leq 0 \text{ when } Z^* \geq t) \end{aligned}$$

Then we can apply the Cauchy-Schwarz inequality to the right-hand side :

$$\begin{aligned} t^2 &\leq \mathbb{E}[(t - Z^*)\mathbb{1}_{\{Z^* < t\}}]^2 \\ &\leq \mathbb{E}[(t - Z^*)^2] \mathbb{E}[\mathbb{1}_{\{Z^* < t\}}^2] \\ &= \mathbb{E}[(t - Z^*)^2] \mathbb{P}(Z^* < t) \\ &= \mathbb{E}[t^2 + (Z^*)^2 - 2tZ^*] \mathbb{P}(Z^* < t) \\ &= (t^2 + \mathbb{V}(Z)) \mathbb{P}(Z^* < t) \quad (\text{since } \mathbb{E}(Z^*) = 0 \text{ and } \mathbb{E}((Z^*)^2) = \mathbb{V}(Z^*) = \mathbb{V}(Z)) \end{aligned}$$

Finally :

$$\begin{aligned}\frac{t^2}{\mathbb{V}(Z) + t^2} &\leq \mathbb{P}(Z - \mathbb{E}(Z) < t) \\ \iff \frac{t^2}{\mathbb{V}(Z) + t^2} &\leq 1 - \mathbb{P}(Z - \mathbb{E}(Z) \geq t) \\ \iff \mathbb{P}(Z - \mathbb{E}(Z) \geq t) &\leq \frac{\mathbb{V}(Z)}{\mathbb{V}(Z) + t^2}\end{aligned}$$

6) Remark : if $m_0 = m_1$, then the bound is true. So we can assume $m_0 \neq m_1$, and, without loss of generality, we can assume $m_0 < m_1$.

We note that we can choose $\Delta_0, \Delta_1 > 0$ such that $m_1 - m_0 = \Delta_0 + \Delta_1$, and consider the classifier $g(m_0 + \Delta_0, 0)$. Then we have :

$$\begin{aligned}L(m_0 + \Delta_0, 0) &= pF_1(\Delta_0 + m_0) + (1 - p)(1 - F_0(\Delta_0 + m_0)) \quad (\text{cf question 1}) \\ &= p\mathbb{P}(X \leq m_0 + \Delta_0 | Y = 1) + (1 - p)\mathbb{P}(X > m_0 + \Delta_0 | Y = 0) \\ &= p\mathbb{P}(\Delta_1 \leq -X - (-m_1) | Y = 1) + (1 - p)\mathbb{P}(X - m_0 > \Delta_0 | Y = 0) \quad (\text{since } m_0 + \Delta_0 = m_1 - \Delta_1) \\ &\leq p \frac{\sigma_1^2}{\sigma_1^2 + \Delta_1^2} + (1 - p) \frac{\sigma_0^2}{\sigma_0^2 + \Delta_0^2} \quad (\text{by Chebyshev-Cantelli inequality}) \\ &\leq \frac{p}{1 + \frac{\Delta_1^2}{\sigma_1^2}} + \frac{1 - p}{1 + \frac{\Delta_0^2}{\sigma_0^2}}\end{aligned}$$

Taking $\Delta_0 = \frac{m_1 - m_0}{\sigma_0 + \sigma_1}$ and $\Delta_1 = \frac{\sigma_1}{\sigma_0} \Delta_0$, we have :

$$L(\Delta_0 + m_0, 0) \leq \frac{p}{1 + \frac{(m_1 - m_0)^2}{(\sigma_0 + \sigma_1)^2}} + \frac{1 - p}{1 + \frac{(m_1 - m_0)^2}{(\sigma_0 + \sigma_1)^2}} = \left(1 + \frac{(m_0 - m_1)^2}{(\sigma_0 + \sigma_1)^2}\right)^{-1}$$

And by definition of L_0 , we have $L_0 \leq L(\Delta_0 + m_0, 0)$, which gives the desired result.

Remark 1 : we can confirm $\Delta_0 + \Delta_1 = \frac{m_1 - m_0}{\sigma_0 + \sigma_1} \sigma_0 + \frac{m_1 - m_0}{\sigma_0 + \sigma_1} \sigma_1 = (m_1 - m_0) \frac{\sigma_0 + \sigma_1}{\sigma_0 + \sigma_1}$.

Remark 2 :

$$\begin{aligned}\frac{1}{1 + \frac{(m_1 - m_0)^2}{(\sigma_0 + \sigma_1)^2}} &\leq \frac{1}{2} \iff 1 + \frac{(m_1 - m_0)^2}{(\sigma_0 + \sigma_1)^2} \geq 2 \\ &\iff \frac{(m_1 - m_0)^2}{(\sigma_0 + \sigma_1)^2} \geq 1 \\ &\iff |m_1 - m_0| \geq \sigma_0 + \sigma_1\end{aligned}$$

7) First let us remind the empirical risk and some results :

$$\begin{aligned}L_n(g) &= \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{Y_i \neq g(X_i)\}} \\ \hat{g}_n &= \arg \min_{g \in \mathcal{G}_0} L_n(g)\end{aligned}$$

$$L(\hat{g}_n) - L_0 \leq 2 \sup_{(x,y) \in \mathbb{R} \times \{0,1\}} |L_n(g) - L(g)|$$

$$\mathbb{P} \left(\sup_{x \in \mathbb{R}} |L_n(g) - L(g)| > \epsilon \right) \leq 2e^{-2n\epsilon^2}, \quad \forall \epsilon > 0, y \in \{0,1\}$$

Then we have :

$$\begin{aligned} \mathbb{P}(L(\hat{g}_n) - L_0 > \epsilon) &\leq \mathbb{P} \left(\sup_{(x,y) \in \mathbb{R} \times \{0,1\}} |L(x, y) - L_n(x, y)| > \frac{\epsilon}{2} \right) \\ &\leq \mathbb{P} \left(\sup_{x \in \mathbb{R}} |L(x, 0) - L_n(x, 0)| > \frac{\epsilon}{2} \right) + \mathbb{P} \left(\sup_{x \in \mathbb{R}} |L_n(x, 1) - L(x, 1)| > \frac{\epsilon}{2} \right) \\ &\leq 4e^{-2n(\epsilon/2)^2} \end{aligned}$$

For any joint distribution of (X, Y) .

Remark : However, this is a relative risk. It also depends on the best risk achievable by the class of classifiers considered (here $\mathcal{G}_0$). If the data is not linearly separable, then L_0 can be high, and the empirical risk minimizer will not perform well (e.g. non-separable data).